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VLMEvalKit: An Open-Source Toolkit for Evaluating Large Multi-Modality Models

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Abstract

We present VLMEvalKit: an open-source toolkit for evaluating large multi-modality models based on PyTorch. The toolkit aims to provide a ***user-friendly*** *and* ***comprehensive*** *framework for researchers and developers to evaluate existing multi-modality models and publish* ***reproducible*** *evaluation results. In VLMEvalKit, we implement over* ***200+ different large multi-modality models****, including both proprietary APIs and open-source models, as well as more than* ***80 different multi-modal benchmarks****. By implementing a single interface, new models can be easily added to the toolkit, while the toolkit automatically handles the remaining workloads, including data preparation, distributed inference, prediction post-processing, and metric calculation. Although the toolkit is currently mainly used for evaluating large vision-language models, its design is compatible with future updates that incorporate additional modalities, such as audio and video. Based on the evaluation results obtained with the toolkit, we host OpenVLM Leaderboard, a comprehensive leaderboard to track the progress of multi-modality learning research. The toolkit is released on GitHub and is actively maintained*[1].

1. Introduction

With the rapid development of Large Language Models (LLMs) [6, 99, 123, 127], Large Multi-Modality Models (LMMs) [100, 121] have also experienced significant advancements. LMMs typically take two or more modalities as input. Most of the research has focused on LMMs for image and text [12, 79], but research has also been extended to other modalities, such as audiotext [23], video [14, 68, 113], or point clouds [156, 159]. Furthermore, there exist LMMs that can simultaneously take more than two modalities as inputs, including proprietary APIs and open-source models [44, 100, 121]. Compared to previous multi-modality models, LMMs, empowered by large language models, exhibit enhanced generalization capability and engage with humans in a variety of conversational styles. These models have not only demonstrated remarkable capabilities in multi-modal perception and reasoning tasks but have also spurred a range of innovative applications.

Quantitative evaluation is crucial in the development of LMMs. As general-purpose models, LMMs must undergo rigorous evaluation in a diverse range of tasks and domains. Comprehensive evaluations not only help users discern the strengths and weaknesses of an LMM, but also offer valuable feedback to developers for ongoing refinement. Unlike 'pre-GPT' models, evaluating LMMs on diversified quantitative benchmarks has become a common practice, both in academic works [78, 165] and commercial APIs [100, 121].

Despite the importance of evaluating LMMs, conducting assessments across dozens of benchmarks can be a daunting task, particularly for small research teams. One must prepare data based on numerous repositories and manage potential environmental conflicts. Moreover, the authors of benchmarks may not provide evaluation results for all LMMs in which users are interested, thus requiring significant effort to compile uncompleted results. To alleviate this challenge, we developed VLMEvalKit, an open-source toolkit designed to facilitate the evaluation of LMMs.

VLMEvalKit aims to provide a comprehensive, user-

[1]VLMEvalKit contributors can join the author list of the report based on their contribution to the repository. Specifically, it requires 3 major contributions (implement a new benchmark, MLLM, or contribute a major feature). We will update the report quarterly and an additional section that details each developer's contribution will be appended in the next update.

Figure 1. VLMEvalKit supports most mainstream commercial APIs including GPT-4o, Gemini, GLM-4v, *etc.*, as well as over 200 open-source LMMs including LLaVA [79], InternVL [19], Idefics [61], *etc.*

friendly evaluation framework for researchers and developers to assess existing Large Multi-Modality Models. Currently, the codebase supports more than 200 different large multi-modality models, spaning proprietary APIs and open-source models (see Fig. 1), and over 80 multi-modal benchmarks covering a wide range of tasks and scenarios (see Tab. 1). The codebase's straightforward design simplifies the integration of new benchmarks or LMMs. Typically, a developer only needs to prepare a single data file or implement a single interface to support new benchmarks or LMMs. Beyond its extensive collection and simplified design, VLMEvalKit significantly eases the work of comprehensive evaluation. Users can launch evaluations across multiple supported LMMs and benchmarks with a single command, generating well-structured evaluation results. The entire process eliminates the need for manual data preparation or post-processing, ensuring a seamless and efficient evaluation experience.

To align with real use cases, VLMEvalKit employs generation-based evaluation across all LMMs and benchmarks. Given their general-purpose nature, LMMs may struggle with adhering to specific formats for solving multiple-choice questions. For a fair comparison, VLMEvalKit utilizes large language models as choice extractors when exact matching fails, thereby mitigating the impact of response styles and enhancing the reliability of evaluation, Leveraging the reliable and reproducible evaluation results provided by VLMEvalKit, we maintain a comprehensive leaderboard to monitor the advancement of LMM development. The leaderboard disseminates valuable insights to the community and has garnered widespread recognition from both academia and industry.

VLMEvalKit is publicly available at `https://github.com/open-compass/VLMEvalKit` under the Apache 2.0 License. The repository includes the complete source codes along with detailed instructions for installation, evaluation, and further development. In subsequent sections, we will delve into the design and features

Benchmark	Keyword	Benchmark	Keyword
MMBench [80]	All-round Capability	CCBench	Chinese Culture
MME [33]	All-round Capability	SEEDBench [64]	All-round Capability
SEEDBench2-Plus [62]	Text-Rich Comprehension	MMT-Bench [167]	Multi-Task
MMMU [170]	Multi-Modal Examination	MMStar [13]	Data Contamination
MathVista [87]	Multi-Modal Math	COCO Caption [16]	Image Caption
OCRBench [82]	Text Understanding	TextVQA [112]	Text Recognition
AI2D [58]	Diagram Understanding	ChartQA [95]	Chart Understanding
DocVQA [97]	Document Understanding	RealWorldQA	Driving Scenarios
HallusionBench [77]	Hallucination & Illusion	POPE [70]	Object Hallucination
MMBench-Video [31]	Video Understanding	OCRVQA [98]	Text Recognition
MLLMGuard [41]	Multi-Modal Safety	AesBench [53]	Aesthetics
MMVet [168]	Subjective Evaluation	LLaVABench [79]	Subjective Evaluation

Table 1. VLMEvalKit supports the evaluation of more than 80 different benchmarks, covering diversified tasks, topics, and scenarios. In the table, we list some benchmarks and the corresponding keywords.

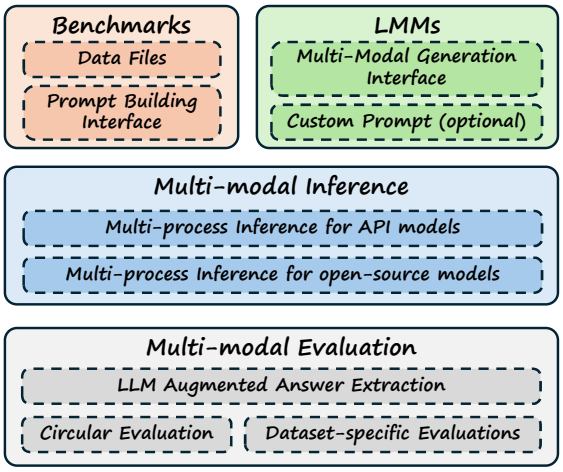

Figure 2. Major Components in VLMEvalKit.

of VLMEvalKit, present and analyze the evaluation results obtained, and discuss potential future developments.

2. VLMEvalKit: Design & Features

Fig. 2 displays the major components in VLMEvalKit. In this section, we will describe each single component in detail.

Benchmarks. VLMEvalKit encompasses dozens of multi-modal benchmarks that span a broad spectrum of tasks and scenarios, as detailed in Tab. 1. The benchmarks are first processed into `.tsv` files, where evaluation samples are stored as individual lines. Each evaluation sample typically includes the `index`, `question`, `answer`, `image` (one or more images encoded in `base64`), and `choices` for multi-choice questions. Each dataset class supports a `.build_prompt()` interface, which the evaluation script utilizes to construct evaluation samples into multi-modal messages. A multi-modal message comprises an interleaved sequence of contents across various modalities. For illustration, consider the following example:

```
[
    dict(type='image', value='path or url of
the image'),
    dict(type='image', value='path or url of
the image'),
    dict(type='text', value='Please list
all the objects that appear in the above
images.')
]
```

Currently, a multi-modal message primarily incorporates `image`, `video` and `text` modalities. The format, however, is extensible and can accommodate additional modalities such as audio or point clouds.

LMMs. VLMEvalKit supports over 200 LMMs, including both commercial APIs and open-source models. A unified `.generate()` interface has been implemented for all LMMs, which accepts a multi-modal message as input and returns the response string. For LMMs that are limited to processing a single image-text pair, the concatenated text messages and the first image are adopted as the default input. To provide flexibility for LMM developers, the `.generate()` interface also includes `dataset_name` as an additional argument. An LMM can optionally implement its own `.build_prompt()` interface to construct custom multi-modal messages, and determine whether to utilize custom multi-modal messages based on the `dataset_name` flag. The unified interface eases the process of comprehensive evaluation. Each newly supported LMM / benchmark can be directly evaluated with all existing benchmarks / LMMs.

Multi-modal Inference. To expedite the multi-modal inference, we support parallelized inference for both commercial APIs, leveraging Python's `multiprocessing`, and open-source models, which are distributed in parallel across multiple GPUs. Additionally, we have implemented a robust inference process so that an interrupted inference process can be resumed with minimal costs of repeated calculation or API calls. For some popular open-source LMMs(like Qwen Series, InternVL Series), we have supported vllm/LMDeploy to accelerate the inference with full utilization of resources.

Multi-modal Evaluation. Predictions from the LMMs will be evaluated based on the specific question format to derive the final metrics. Benchmarks supported in VLMEvalKit can be categorized into three primary types: 1. Multi-choice questions (MCQ), where the model must select from given options and respond with the corresponding label (*e.g.*, A, B); 2. Yes-or-No questions (Y/N), requiring a straightforward 'Yes' or 'No' answer; 3. Open-ended questions, which necessitate a free-form response. Notably, a significant number of LMMs struggle to adhere to the instructions precisely, often producing responses that are not well-formatted for MCQ and Y/N benchmarks. To improve the precision of evaluations and counteract the influence of varied response styles, VLMEvalKit offers the option to integrate LLM-augmented answer extraction specifically for MCQ and Y/N benchmarks. We first adopt exact matching to match the response with the option labels or contents. Should this step fail, the toolkit then prompts an LLM (such as ChatGPT) to match the response with the option that most closely aligns with semantic meaning. This strategy helps us better understand the real performance of LMMs, particularly for commercial APIs [80].

Another notable challenge in assessing MCQ benchmarks is the inherent **variance**. When employing random guessing, an LMM may correctly answer $1/N$ questions for N-option multi-choice benchmarks. Besides, we find that the outcome of an LMM can be significantly influenced by the order of options. These factors make the evaluation results highly variable and the performance gap between LMMs less discernible. To address this, VLMEvalKit offers an option to evaluate all MCQ benchmarks in **Circular** mode: N options of a MCQ will be shifted in circular N times to formulate N new questions. The results count only if an LMM accurately answers all N circular-shifted MCQs. The CircularEval strategy can more effectively assess the real comprehension of an LMM on MCQ benchmarks, allowing users to identify more pronounced performance disparities between models.

For open-ended benchmarks, VLMEvalKit follows the original practice for conducting the evaluation. Subjective benchmarks like MMVet [168] or LLaVABench [79] adopts GPT-4 for marking, based on the semantic similarity between LMM responses and the reference answer. For VQA benchmarks [98, 112], we follow the standard practices to calculate accuracies based on heuristic matching.

3. Evaluation Results

3.1. General VQA Benchmarks

Utilizing VLMEvalKit, one can conduct comprehensive evaluations of an LMM across numerous benchmarks to gain a thorough understanding of its strengths and weaknesses. We publish all evaluation results on OpenVLM Leaderboard. Our core leaderboard is based on evaluations from eight distinct benchmarks: 1. MMBench v1.1 [test] [80][2] (all-round capability); 2. MMStar [13] (data contamination); 3. MMMU [170] [val] (multi-modal examination); 4. MathVista [mini-test] [87] (multi-modal math); 5. HallusionBench [77] (hallucination & illusion); 6. AI2D [test] [58] (diagram understanding); 7. OCRBench [82] (text understanding); 8. MMVet [168] (subjective evaluation). The selection encompasses a diverse array of tasks, and the average score across these benchmarks serves as a reliable indicator of the general capabilities of LMMs. In

[2] We report the average score of English and Chinese test splits.

Method	Param.	Avg. Score	MMBenchV1.1	MMStar	MMMU	Math.	Hallu.	AI2D	OCR.	MMVet
CongRong-v2.0 [24]	N/A	80.7	88.1	**75.3**	**75.6**	76.8	63.2	**90.0**	**927**	83.9
SenseNova-V6-Pro [108]	N/A	80.4	88.0	73.7	70.4	76.9	**67.1**	89.2	895	**88.2**
Gemini-2.5-Pro [121]	N/A	80.1	**88.3**	73.6	74.7	**80.9**	64.1	89.5	862	83.3
InternVL3-78B [189]	78B	79.1	87.7	73.4	72.2	79.0	59.1	89.8	908	80.7
InternVL3-38B [189]	38B	77.8	86.8	72.6	69.7	76.3	58.4	88.7	886	81.1
Step-1o [115]	N/A	77.7	87.3	69.3	69.9	74.7	55.8	89.1	926	82.8
SenseNova [108]	N/A	77.4	85.7	72.7	69.6	78.4	57.4	87.8	894	78.2
InternVL2.5-78B-MPO [18]	78B	77	87.7	72.1	68.2	76.6	58.1	89.2	909	73.5
GLM-4v-plus-20250111 [5]	N/A	76.7	85.9	72.5	69.9	73.5	58.5	86.7	908	75.7
Ovis2-34B [89]	34B	76.5	86.5	69.2	66.7	76.1	58.8	88.3	894	77.7

Table 2. The evaluation results of LMMs on general VQA benchmarks. The table displays the top-10 LMMs, including both commercial APIs and open-source LMMs(**till 2025.06.20**), in the descending order of average score. When calculating the average score, scores of each benchmark are normalized to the range of 0 to 100. When reporting the parameter size, '1B' means 10^9 parameters. Commercial APIs are denoted with blue background . **Bold** and underline indicates the best and second-best performance in each group.

Method	Param.	Avg. Score	MathVista	MathVision	MathVerse	DynaMath	WeMath	LogicVista
Seed1.5-VL [42]	N/A	73.3	**86.8**	67.3	**79.3**	56.1	77.5	72.7
Gemini-2.5-Pro [121]	N/A	72.5	80.9	**69.1**	76.9	**56.3**	**78**	**73.8**
Doubao-1.5-pro [120]	N/A	61.6	78.6	51.5	64.7	44.9	65.7	64.2
Gemini-2.0-Pro [121]	N/A	56.6	71.3	48.1	67.3	43.3	56.5	53.2
ChatGPT-4o-latest [99]	N/A	54.8	71.6	43.8	49.9	48.5	50.6	64.4
GPT-4.1-20250414 [99]	N/A	54	70.4	45.1	48.9	43.3	55.5	61.1
InternVL3-78B [99]	78B	51	79	38.8	51	35.1	46.1	55.9
Gemini-2.0-flash [121]	N/A	50.6	70.4	43.6	47.8	42.1	47.4	52.3
Claude-3.7-sonnet-20250219 [6]	N/A	50.4	66.8	41.9	46.7	39.7	49.3	58.2
InternVL3-38B [189]	38B	50.3	76.3	35.9	48.2	34.7	49.3	57.7

Table 3. The evaluation results of LMMs on image reasoning benchmarks. The table displays the top-10 LMMs, including both commercial APIs and open-source LMMs(**till 2025.02.28**), in the descending order of average score.

Tab. 2, we present the performance of top-10 commercial APIs and the top-10 open-source models evaluated on this suite of benchmarks.

As illustrated in the table, after nearly two years of development, the performance gap between commercial APIs and open-source LMMs on general VQA benchmarks has significantly narrowed. Open-source LMMs now demonstrate strong capabilities in general understanding tasks, often matching or even surpassing the performance of commercial APIs. Among the top-10 models, four are commercial APIs, while the other five are open-source models. Although the top three models, CongRong-v2.0 [24], SenseNova-V6-Pro [108], and Gemini-2.5-Pro [121] remain commercial APIs, their performance is remarkably close to that of the fourth-ranked model, InternVL3-78B [189]. From the first-ranked model, CongRong-v2.0, to the tenth-ranked model, Ovis2-34B, the average score decreases by only 4.2% (from 80.7% to 76.5%). Notably, prominent commercial series such as GPT [99] and Claude [6] fail to rank within the top 10, whereas open-source models like the InternVL and Ovis series exhibit exceptional performance and competitive strength.

3.2. Image Reasoning Benchmarks

With the rapid advancement of LMMs, reasoning capabilities have garnered increasing attention. To systematically assess these capabilities, we have compiled several widely recognized multi-modal benchmarks designed for reasoning tasks, including MathVista [87], MathVision [136], MathVerse [176], DynaMath [190], WeMath [105], and LogicVista [154]. Evaluations were conducted on OpenVLM Reasoning Leaderboard. The selected benchmarks primarily emphasize mathematical and puzzle-solving problems, spanning areas such as geometry, alge-

Method	Param.	Frame.	Avg. Score	MVBench	Video-MME(w/o subs)	MMBench-Video	TempCompass	MLVU
InternVL3-78B [189]	78.4	64	**72.7**	**79.2**	**73.1**	1.81	77.0	**74.0**
InternVL3-38B [189]	38.4	64	71.9	76.0	72.1	1.80	78.5	72.7
Qwen2.5-VL-72B [7]	73.4	64	68.6	71.3	68.6	1.84	77.7	64.1
InternVL2.5-78B [18]	78.4	64	68.1	75.6	72.1	1.98	56.0	70.9
Qwen2-VL-72B [137]	73.4	64	67.7	70.2	67.3	1.70	**79.4**	65.2
InternVL3-8B [189]	7.94	64	66.8	73.2	66.0	1.69	70.4	68.0
LLaVA-Video-72B-Qwen2 [179]	7.94	64	66.2	63.1	70.5	1.71	70.4	70.0
Aria [66]	25.3	64	66.2	67.9	66.0	1.81	69.6	67.1
Gemini-2.0-flash [121]	N/A	64	64.6	59.4	71.0	**2.01**	57.9	68.0
GPT-4o-20240806 [99]	N/A	16	63.1	57.5	67.9	1.87	72.7	54.9
Qwen2.5-Omni-7B [155]	10.7	64	64.5	69.0	64.1	1.65	70.7	63.6

Table 4. The evaluation results of LMMs on video understanding benchmarks. The table displays the top-10 LMMs, including both commercial APIs and open-source LMMs(**till 2025.06.25**), in the descending order of average score. Due to resource limitations, we only tested a few commercial APIs.

bra, logic, and related domains. The performance metrics of the top-10 models are summarized in Tab. 3.

In multi-modal reasoning tasks, a significant performance gap remains evident between commercial APIs and open-source LMMs. Generally, commercial APIs demonstrate a marked advantage over their open-source counterparts. Notably, the top six performing LMMs are all proprietary APIs. InternVL3-78B [189], the highest-performing open-source LMM on reasoning benchmarks, achieves an average score of only 51, highlighting a considerable gap compared to the leading model Seed1.5-VL [42](73.3). Further analysis reveals that most high-performing models achieve accuracy exceeding 85% on MathVista [87]. In contrast, only Seed1.5-VL and Gemini-2.5-Pro achieve accuracy greater than 50% on DynaMath [190], underscoring the challenges posed by this benchmark.

3.3. Video Benchmarks

Video has become essential in daily life, driving communication, learning, and entertainment. With the rapid rise of video content and progress in LMM capability, video understanding ability is increasingly vital for LMMs. To fully examine the video understanding capability, we have added support for some popular common video understanding benchmarks, including MMBench-Video [31], Video-MME [34], MLVU [188], TempCompass [81], MVBench [68], and other video datasets. Besides, in order to have a glimpse into the overall video understanding performance of LMMs, we also selected some representative video benchmarks and evaluated over forty LMMs, with the results presented on the OpenVLM Video Leaderboard. The selected benchmarks primarily cover various question types (multiple choice, open-ended questions, true/false questions, etc.) and various video lengths, mainly examining the model's general video understanding ability. The performance metrics of the top-10 models are summarized in Tab. 4.

Due to the limited resources of commercial APIs and the significant costs associated with video understanding evaluations, we focused on testing the video understanding capabilities of open-source LMMs. It's easy to observe that InternVL and Qwen Series demonstrate remarkable performance in video understanding benchmarks. InternVL3-78B [189] achieves the highest average score of 72.7, and Qwen2.5 lags slightly behind them, mainly revealed in Video-MME [34] and MLVU [188]. With a 64-frame input window, Gemini-2.0-flash [121] still lags behind the state of the art on several tasks, possibly because it refuses to answer some questions. GPT-4o [99] achieves 63.1 with only 16 frames, showing that commercial APIs can still compete in video understanding even under a short context.

4. Discussion

We have released VLMEvalKit, an open-source toolkit designed for the evaluation of large multi-modality models. VLMEvalKit encompasses a comprehensive collection of over 300 LMMs and more than 100 multi-modal benchmarks. The codebase is structured with simplicity in mind, facilitating the easy integration of new LMMs or benchmarks. Our framework is thoughtfully designed to extend its capabilities beyond the vision modality and incorporate other modalities like audio. Moving forward, the development of VLMEvalKit will focus on expanding the repertoire of LMMs and benchmarks for video and other modalities. We are optimistic that this repository, along with all released resources, will contribute to advancing research in multi-modal learning.

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A. Benchmarks

In this section, we list all supported benchmarks with a brief introduction and categorize them into multiple categories.

A.1. General Capability

- **MMBench Series** [80]: The MMBench Series, which includes MMBench and its Chinese version MMBench-CN, serves as a pioneering benchmark for assessing Large Vision-Language Models (LVLMs) across 20 distinct capabilities. MMBench-CN adapts the questions and choices of MMBench into Chinese based on GPT-4. These benchmarks are designed to promote a more precise and comprehensive evaluation of VLMs.
- **SEEDBench & SEEDBench2** [64]: SEEDBench comprises 19,000 multiple-choice questions that span 12 dimensions, while SEEDBench2 expands this dataset to 24,000 questions across 27 dimensions. Both benchmarks provide a robust framework for evaluating the multifaceted capabilities of AI models.
- **MME** [33]: a holistic benchmark aimed at evaluating MLLMs. It assesses perception and cognition across 14 subtasks, featuring manually curated instruction-answer pairs to prevent data leakage and ensure a fair comparison without the need for prompt engineering.
- **MMT-Bench** [167]: a comprehensive benchmark designed to assess LVLMs across massive multimodal tasks requiring expert knowledge and deliberate visual recognition, localization, reasoning, and planning.
- **MME-RealWorld** [182]: a large-scale benchmark for evaluating MLLMs in real-world scenarios. It comprises 29,429 question-answer pairs and encompasses 43 subtasks, providing a rich dataset for testing the practical applications of MLLMs.

A.2. Text Recognition & Understanding

- **TextVQA** [112]: a dataset for assessing visual reasoning based on textual content within images. TextVQA challenges models to read and reason about the text present in images to answer corresponding questions. To accomplish this, models must integrate the textual information found in the images and utilize it to formulate responses to TextVQA's queries.
- **OCRVQA** [98]: a benchmark for visual question answering by reading text in images. It introduces the OCR-VQA-200K dataset, which consists of over 200,000 book cover images. This study integrates techniques from Optical Character Recognition (OCR) and Visual Question Answering (VQA) to tackle the novel challenges associated with this dataset.
- **OCRBench** [82]: a comprehensive evaluation benchmark designed to assess the Optical Character Recognition (OCR) capabilities of LMMs across various text-related visual tasks, including text recognition, scene text-centric VQA, document-oriented VQA, key information extraction, and handwritten mathematical expression recognition.
- **DocVQA** [97]: designed to encourage a "purpose-driven" point of view in Document Analysis and Recognition research, where the document content is extracted and used to respond to high-level tasks defined by the human consumers of this information.

A.3. Structuralized Content Understanding

- **ChartQA** [95]: a benchmark designed to assess question answering skills related to charts, with an emphasis on complex visual and logical reasoning. The dataset comprises 9,600 human-written questions and 23,100 machine-generated questions based on chart summaries. It challenges models to perform intricate reasoning that includes logical and arithmetic operations as well as to analyze visual chart features.
- **InfoVQA** [96]: a dataset aimed at enhancing the automatic understanding of infographic images through VQA. It includes a diverse collection of infographics paired with natural language questions and answers, focusing on the need for models to reason across document layouts, textual content, graphical elements, and data visualizations.
- **TableVQA-Bench** [59]: a benchmark created for visual question answering focused on table data. It integrates images and question-answer pairs that were not previously available in existing datasets. The benchmark consists of 1,500 QA pairs generated from text-formatted tables using a large language model (LLM).
- **SEEDBench2-Plus** [62]: a benchmark that includes 2,300 visual questions covering three categories (Charts, Maps, and Webs) and 63 fine-grained data types found in real-world scenarios. It is designed to evaluate the capabilities of multimodal large language models in understanding text-rich visual content.

A.4. Subjective (LLM Judge)

- **MM-Vet** [168]: an evaluation benchmark for LMMs designed to assess their performance on complex multimodal tasks. It provides a systematic structure for these tasks and proposes unified evaluation metrics. The benchmark utilizes Large Language Model (LLM)-based evaluators to assess a variety of question and answer types, ensuring a comprehensive analysis of LMM capabilities.
- **LLaVABench** [79]: a dataset created to evaluate the capabilities of LMMs in more challenging tasks and their ability to generalize to new domains. It con-

sists of a diverse set of 24 images with 60 questions in total, including indoor and outdoor scenes, memes, paintings, sketches, etc., and each image with a highly-detailed and manually-curated description and a proper selection of questions.

A.5. Mathematics & Examination

- **MathVista** [87]: a benchmark designed to evaluate the mathematical reasoning capabilities of LLMs and LMMs in visual contexts. It comprises 6,141 examples from 28 existing multimodal datasets and three new datasets (IQTest, FunctionQA, and PaperQA), challenging models with deep visual understanding and compositional reasoning.
- **MathVision** [87]: a dataset addressing limitations in existing benchmarks for evaluating mathematical reasoning in visual contexts. It includes 3,040 high-quality mathematical problems sourced from actual competitions, spanning 16 disciplines and varying in difficulty across five levels.
- **ScienceQA-IMG** [88]: a multimodal benchmark for answering science-related questions. It contains approximately 21,000 questions and annotations, promoting reasoning through the Chain of Thought (CoT) approach.
- **MMMU** [170]: a comprehensive benchmark for evaluating multimodal models on tasks that require expert-level knowledge. It features 11,500 questions across six disciplines and 183 subfields.

A.6. Low-level & Aesthetics

- **AesBench** [53]: a benchmark for evaluating aesthetic perception capabilities of MLLMs with an expert-labeled database and from four shallow-to-deep perspectives.
- **Q-Bench** [149]: a benchmark for evaluating MLLMs in low-level vision tasks, including perception, description, and assessment. Only multi-choice questions are included in our implementation.
- **A-Bench** [184]: a benchmark for evaluating LMMs in assessing AI-generated images, including 2864 AIGIs from 16 text-to-images models.

A.7. Long-Context, Multi-Image & Multi-Turn

- **DUDE** [128]: a comprehensive benchmark aimed at advancing Document AI by addressing limitations in current methods for visually-rich documents. The benchmarks include 6315 visual questions, with an average of 5.7 images per question.
- **MMLongBench-Doc** [92]: a benchmark designed to evaluate understanding of long-context documents in LVLMs. It consists of 1,091 expert-annotated questions based on 135 lengthy PDF documents, averaging 47.5 pages and 21,214 tokens each. This benchmark uniquely requires that answers be derived from various multimodal sources, including text, images, charts, tables, and layout structures, and 33% of the questions require cross-page evidence.
- **SlideVQA** [117]: a multi-image document VQA dataset, consisting of 52k+ slide images from 2.6k+ slide decks and 14.5k questions. It requires complex reasoning, including numerical and multihop reasoning, and provides annotated arithmetic expressions for numerical answers.
- **BLINK** [38]: a benchmark with 14 challenging visual perception tasks for multimodal LLMs, with around 60% of visual questions consisting of multiple images as inputs.
- **MuirBench** [131]: a comprehensive benchmark for multimodal LLMs on multi-image understanding with 12 tasks, 10 relation categories, 11264 images and 2600 questions.
- **MMDU** [83]: a benchmark designed to evaluate VLMs capabilities in multi-turn, multi-image dialogues, addressing their limitations in handling real-world, complex conversational scenarios. MMDU features longer dialogues with up to 27 turns and 20 images, challenging current LVLMs.

A.8. Hallucination

- **HallusionBench** [77]: a diagnostic benchmark for evaluating large visual-language models' reasoning about image-context relationships, focusing on visual illusions and language hallucinations. Consists of 346 images and 1,129 questions to assess response consistency, failure modes, and logical reasoning.
- **POPE** [70]: a benchmark for evaluating object hallucination with three tracks (random, popular, and adversarial) and a total of $\sim$ 9k cases.

A.9. Multilingual

- **MMMB and Multilingual MMBench** [116]: MMMB is a multilingual multimodal benchmark with six languages, 15 categories, and 12k questions. Multilingual MMBench extends MMBench-DEV to six languages.
- **MT-VQA** [118]: a benchmark designed for multilingual Text-Centric Visual Question Answering, addressing limitations of previous datasets that focus on high-resource languages and rely on translation-based approaches. The work introduces high-quality human annotations across nine diverse languages, avoiding "visual-textual misalignment" and improving multilingual scene understanding.

A.10. Miscellaneous

- **MLLMGuard (Safety)** [41]: a multi-dimensional safety evaluation suite for MLLMs, including a bilingual image-text evaluation dataset, inference utilities, and a set of lightweight evaluators.
- **GMAI-MMBench (Medical)** [15]: a comprehensive multimodal evaluation benchmark for VLMs in the medical field across multiple datasets, image modalities, and clinical tasks.
- **RealWorldQA (Autonomous Driving)** [153]: a benchmark for evaluating real-world spatial understanding of multimodal AI models with over 700 images from various scenarios.
- **COCO Caption (Captioning)** [76]: Contains 5,000 samples from the COCO Caption Validation set and prompts VLMs to describe the images.
- **MMStar (Data Contamination)** [13]: a vision-indispensable multi-modal benchmark addressing issues of unnecessary visual content and data leakage in evaluating LVLMs. Comprises 1,500 curated samples and proposes new metrics for data leakage.
- **TaskMeAnything ImageQA** [174]: a benchmark generation engine for creating tailored benchmarks with an extendable taxonomy of visual assets.
- **A-OKVQA** [107]: a benchmark for assessing VQA using world knowledge and commonsense reasoning with around 25k crowdsourced questions.
- **VCR-wiki** [177]: Visual Caption Restoration (VCR) is a new vision-language task that requires models to restore partially obscured texts in images using pixel-level hints. Unlike traditional tasks that often rely on optical character recognition or masked language modeling, VCR emphasizes the integration of visual, textual, and contextual cues to achieve accurate text restoration.

A.11. Video Understanding

- **MMBench-Video** [31]: a long-form, multi-shot benchmark with about 600 YouTube videos (30 s–6 min) drawn from 16 everyday domains. It offers roughly 2000 volunteer-written Q&A pairs that probe 26 fine-grained skills and uses a GPT-4 adjudication pipeline for reliable scoring, making it a concise testbed for holistic video understanding.
- **Video-MME** [34]: the first "full-spectrum" evaluation for video LLMs, containing 900 clips plus 2700 human-annotated Q&A pairs. Videos span short to hour-long segments across six visual domains, and the benchmark explicitly mixes frames, audio, and subtitles to examine cross-modal reasoning over varied temporal scales.
- **MVBench** [68]: targets temporal reasoning by converting 20 classic vision tasks into dynamic versions that cannot be solved with a single frame. Automatic conversion of public annotations into multiple-choice QAs ensures fairness, and early results show mainstream MLLMs still fall short on many perception-to-cognition temporal skills.
- **MLVU** [188]: focuses on long-video comprehension, gathering clips from 3 min up to 2 h and organising nine tasks that demand both global narrative understanding and fine-grained detail tracking. Initial evaluations (20 models, incl. GPT-4o) reveal that even the best system scores only 64.6%, underscoring the difficulty of sustained reasoning over extended contexts.
- **TempCompass** [81]: an benchmark designed to gauge temporal perception. It covers diverse temporal aspects (action granularity, motion speed, event order, attribute change, etc.) and multiple task formats (multi-choice, yes/no, caption matching/generation). Carefully crafted "conflicting videos" reduce single-frame shortcuts, offering a stricter measure of whether models truly exploit temporal cues.

B. Acknowledgements

We make our best effort to list all known contributions to the repository. If some of your contributions are missing, please contact opencompass@pjlab.org.cn.

B.1. Contributors with 3+ Major Contributions

We list contributors who have made 3+ significant contributions to the development of VLMEvalKit.

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